MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.

Each subject is divided into 2 sections:

Section 1: Single Correct Type Questions (4 options, only one correct)

Section 2: Numerical Value Type Questions (answer to be entered as numerical value)

Marking Scheme:

Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.

Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

In a photoelectric experiment, with light of wavelength λ, the fastest electron has speed v. If the exciting wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electron will become

(A) $v\sqrt{\frac{3}{4}}$

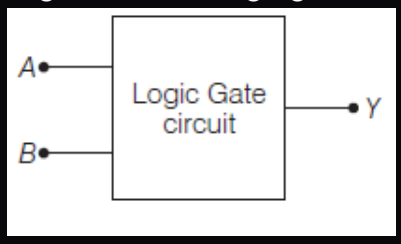
(B) $v\sqrt{\frac{4}{3}}$

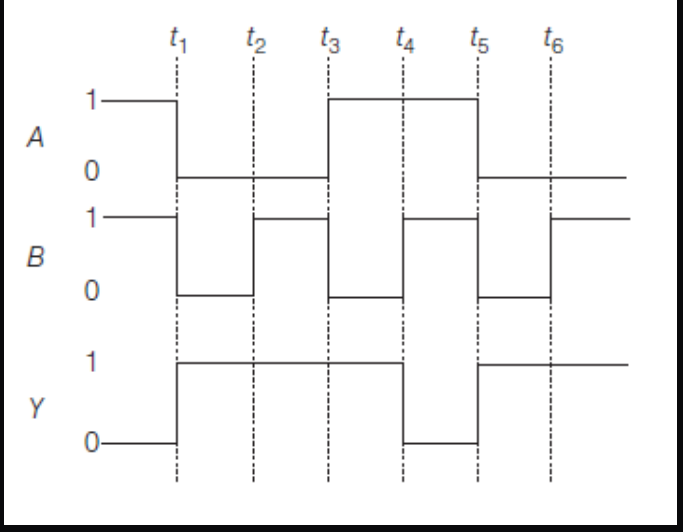
(C) less than $v\sqrt{\frac{3}{4}}$

(D) greater than $v\sqrt{\frac{4}{3}}$

Q.2

The following figure shows a logic gate circuit with inputs A and B and output Y. If the voltage waveforms of A, B and Y are as given, then the logic gate is





(A) NOR gate

(B) OR gate

(C) AND gate

(D) NAND gate

Q.3

In an α-decay the Kinetic energy of α particle is 48MeV and Q-value of the reaction is 50MeV. The mass number of the mother nucleus is:- (Assume that daughter nucleus is in ground state)

(A) 96

(B) 100

(C) 104

(D) none of these

Q.4

For a Young's double slit experiment by coherent light source the intensity at central maxima was found to be I_0. The intensity of light formed at a distance of $\beta/3$ from central maxima is (where ' β ' is the fringe width)

(A) $\frac{I_0}{4}$

(B) $\frac{3I_0}{4}$

(C) $\frac{I_0}{2}$

(D) $\frac{I_0\sqrt{3}}{2}$

Q.5

A battery of internal resistance 2Ω is connected to a variable resistor whose value can vary from 4Ω to 10 Ω. The resistance is initially set at 4Ω. If the resistance is now increased then :

(A) power consumed by it will decrease

(B) power consumed by it will increase

(C) power consumed by it may increase or may decrease

(D) power consumed will first increase then decrease.

Q.6

Two block of masses m_1 and m_2 are connected with a massless unstretched spring and placed over a plank moving with an acceleration ' a ' as shown in figure. The coefficient of friction between the blocks and platform is μ.



(A) spring will be stretched if $\mathrm{a} > \mu \mathrm{g}$

(B) spring will be compressed if $\mathrm{a} \leq \mu \mathrm{g}$

(C) spring will neither be compressed nor be stretched for a $\leq \mu \mathrm{g}$

(D) spring will be in its natural length under all conditions

Q.7

The escape velocity for the earth is Ves. Find out escape velocity for a planet whose radius is four times and density is nine times that of the earth. If is $n \times 6$ Ves. Find n.

(A) 6

(B) 4

(C) 8

(D) 2

Q.8

In the hydrogen atom, an electron makes a transition from $n = 2$ to $n = 1$. The magnetic field produced by the circulating electron at the nucleus

(A) decreases 16 times

(B) increases 4 times

(C) decreases 4 times

(D) increases 32 times

Q.9

The secondary coil of an ideal step down transformer is delivering 500 watt power at 12.5 A current. If the ratio of turns in the primary to the secondary is $5 : 1$, then the current flowing in the primary coil will be :

(A) 62.5 A

(B) 2.5 A

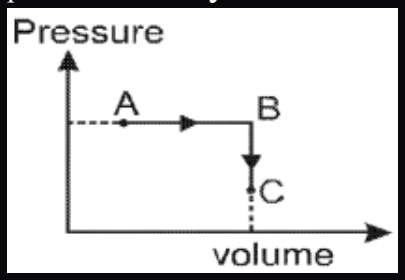
(C) 6 A

(D) 0.4 A

Q.10

In the process ABC for an ideal mono-atomic gas, the temperature at states A and C are equal. The heat released in the

process BC is Q. The work done in the process A to B is equal to (AB is isobaric and BC is isochoric process) :



(A) Q

(B) $\frac{Q}{2}$

(C) $\frac{2Q}{3}$

(D) $\frac{3Q}{2}$

Q.11

A body is thrown horizontally with a velocity $\sqrt{2gh}$ from the top of a tower of height h. It strikes the level ground through the foot of the tower at a distance x from the tower. The value of x is:

(A) h

(B) $\frac{h}{2}$

(C) $2\ \mathrm{h}$

(D) $\frac{2\ \mathrm{h}}{3}$

Q.12

In the relation
$P = \frac{\alpha}{\beta} e^{\frac{\alpha z}{k\theta}}$
P is pressure, Z is distance, k is Boltzmann constant and θ is the temperature. The dimensional formula of β will be :

(A) $\left[\mathrm{M}^0\ \mathrm{L}^2\ \mathrm{T}^0\right]$

(B) $\left[M' L^2 T'\right]$

(C) $\left[M^1 L^0 T^{-1}\right]$

(D) $\left[M^0 L^2 T^{-1}\right]$

Q.13

In a process the density of a gas remains constant. If the temperature is doubled, then the change in the pressure will be :

(A) 100% increase

(B) 200% increase

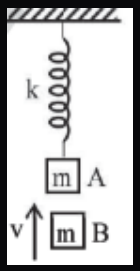
(C) 50% decrease

(D) 25% decrease

Q.14

Block ' A ' is hanging from a vertical spring and is at rest. Block ' B ' strikes the block ' A ' with velocity ' v ' and sticks to it.

Then the value of ' v ' for which the spring just attains natural length is -



(A) $\sqrt{\frac{60mg^2}{k}}$

(B) $\sqrt{\frac{6mg^2}{k}}$

(C) $\sqrt{\frac{10mg^2}{k}}$

(D) None of these

Q.15

How many significant numbers are there in 4.150×10^{-4} ?

(A) 1

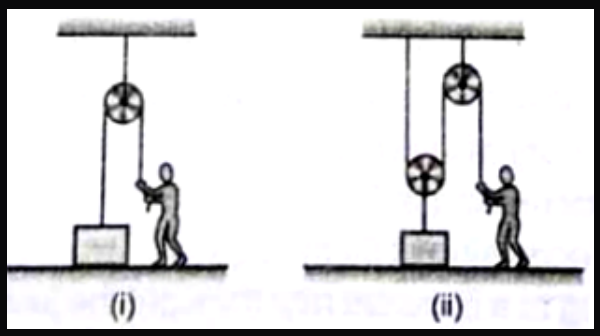
(B) 2

(C) 3

(D) 4

Q.16

In the figure shown, a person wants to raise a block lying on the ground to a height h. In both the cases if time required is same then in which case he has to exert more force. Assume pulleys and strings light.



(A) (i)

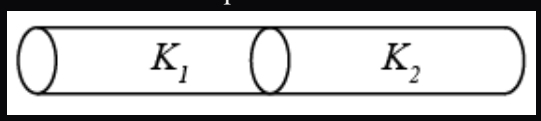
(B) (ii)

(C) same in both

(D) Cannot be determined

Q.17

Statement-1: Two solid cylindrical rods of identical size and different thermal conductivity K_1 and K_2 are connected in series. Then the equivalent thermal conductivity of two rod system is less than the value of thermal conductivity of either rod.



Statement-2 : For two cylindrical rods of identical size and different thermal conductivity K_1 and K_2 connected in series, the equivalent thermal conductivity K is given by

$\frac{2}{\mathrm{K}}=\frac{1}{\mathrm{K}_1}+\frac{1}{\mathrm{K}_2}$

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

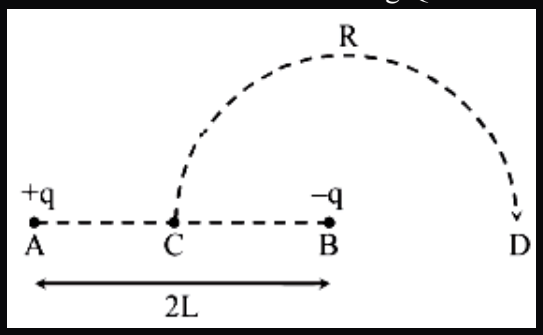
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True.

Q.18

Charges $+\mathrm{q}$ and $-\mathrm{q}$ are placed at A and B respectively which are at a distance $2\ \mathrm{L}$ apart. If C is the midpoint between A and B then work done in moving Q from C to D (through semicircular path CRD) and from C to ∞ are respectively -



(A) $\frac{\mathrm{qQ}}{2\pi\varepsilon_0\,\mathrm{L}}$ and infinity

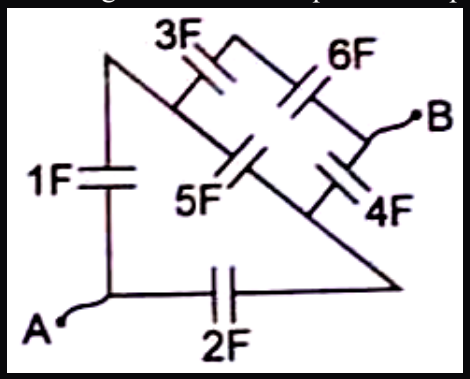
(B) $\frac{\mathrm{qQ}}{6\pi\varepsilon_0\,\mathrm{L}}$ and zero

(C) zero, zero

(D) $\frac{-qQ}{6\pi\varepsilon_0\,\mathrm{L}}$ and zero

Q.19

In the figure shown the equivalent capacitance between ' A ' and ' B ' is :



(A) $3.75\ \mathrm{F}$

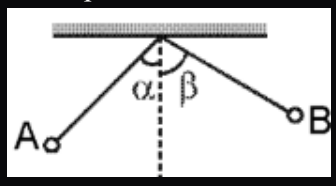
(B) $2\ \mathrm{F}$

(C) $21\ \mathrm{F}$

(D) $16\ \mathrm{F}$

Q.20

Two identical simple pendulums A and B have same point of suspension, having length ℓ each. They are displaced by an angle α and β (α and β are very small and $\beta > \alpha$) and released from rest. Find the time after which B reaches at its initial position for the first time. Assume collision to be elastic and both pendulums move in same plane.



(A) $\pi\sqrt{\frac{\ell}{g}}$

(B) $2\pi\sqrt{\frac{\ell}{g}}$

(C) $\frac{\pi\beta}{\alpha}\sqrt{\frac{\ell}{g}}$

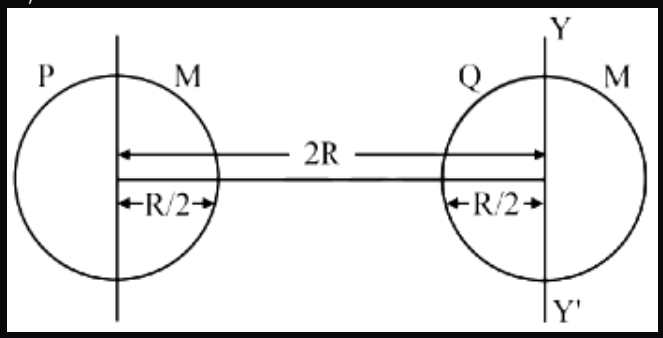
(D) $\frac{2\pi\beta}{\alpha}\sqrt{\frac{\ell}{g}}$

Q.21

A tuning fork and uniform string of length $100\ \mathrm{cm}$ give 4 beats $/\mathrm{s}$. The string is made shorter without any change in tension and the mode of oscillation, until its frequency becomes equal to that of the fork. If now the length of the string is $99\ \mathrm{cm}$, and the frequency of the fork is $100\mathrm{nHz}$, find the value of n.

Q.22

Two spheres each of mass M and radius $R/2$ are connected with a massless rod of length $2R$ as shown in the figure. The moment of inertia of the system about an axis passing through the centre of one of the spheres are perpendicular to the rod is $k/5MR^2$, find k.



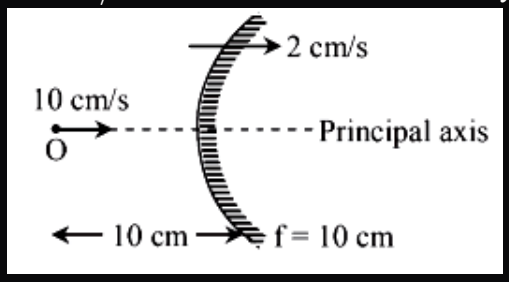
Q.23

From Meter Bridge, resistivity of a wire comes from
$\rho = \frac{\pi D^2 s}{4L}\frac{\ell}{100-\ell}$
where ℓ is balance length, D is diameter of wire, S is resistance, L is total length of wire. Find the value of ℓ (in cm) corresponding to max. permissible error in ρ.

Q.24

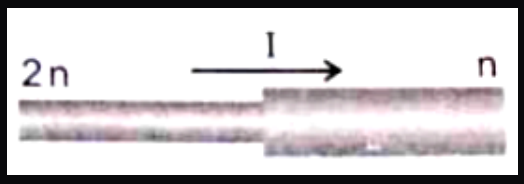
What is the velocity (in cm/s) of image in situtation shown below. (O-object f = focal length). Object moves with velocity

$10\ \mathrm{cm/s}$ and mirror moves with velocity $2\ \mathrm{cm/s}$ as shown.



Q.25

Two cylindrical rods of uniform cross-section area A and $2A$, having free electrons per unit volume $2n$ and n respectively are joined in series. A current I flows through them in steady state. Then the ratio of drift velocity of free electron in left rod to drift velocity of electron in the right rod is $\left(\frac{v_L}{V_R}\right)$ is :



Q.26

Identify the correct order of ionic radii of the given species.

(A) $\mathrm{Mn^{7+} > V^{5+} > Sc^{3+} > K^{+} > S^{2-}}$

(B) $\mathrm{S^{2-} > K^{+} > Sc^{3+} > V^{5+} > Mn^{7+}}$

(C) $\mathrm{S^{2-} > K^{+} > V^{5+} > Sc^{3+} > Mn^{7+}}$

(D) $\mathrm{K^{+} > S^{2-} > Sc^{3+} > V^{5+} > Mn^{7+}}$

Q.27

The different types of bonds present in $\mathrm{CuSO_4 \cdot 5H_2O}$ are

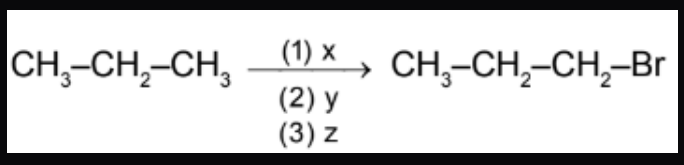
(A) only ionic and covalent

(B) only ionic, covalent and coordinate covalent

(C) only covalent, coordinate covalent and hydrogen

(D) ionic, covalent, coordinate covalent and hydrogen

Q.28



x, y, z are respectively :

(A) $\mathrm{Br_2/h\nu}$, conc. $\mathrm{H_2SO_4}$, $\mathrm{ThO_2/\Delta}$

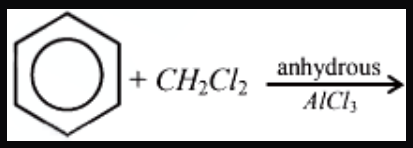
(B) $\mathrm{Cl_2/h\nu}$, alc. $\mathrm{KOH/\Delta}$, $\mathrm{HBr/R_2O_2}$

(C) $Cl_2/h\nu, tBuO^{\ominus}/\Delta, HBr$ (dark)

(D) $F_2/hv, Al_2O_3/\Delta$, NBS

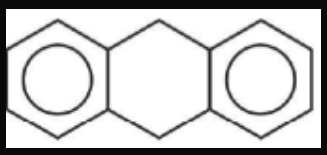
Q.29

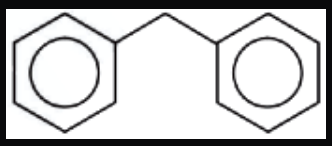
The reaction of



gives :-

(A) 

(B) 

(C) 

(D) All of these

Q.30

If the vapour pressures of pure A and pure B at $298\ K$ are 60 and 15 torr respectively, what woule be the mole percent of A in vapour phase (at this temperature) in a solution that contains 20 mole per cent of A in the $(A + B)$ binary mixture in the liquid phase ? Express your answer as $0.x$. What is x

(A) 3

(B) 4

(C) 5

(D) 6

Q.31

Which of the following statements is correct about "shielding effect"?

1. It is effective, when the orbitals in the inner shells are completely filled.
2. Inner shells do not show any effect.
3. With increase of shielding down the group, ionisation energy increases.
4. With increasing nuclear charge, the shielding effect increases.

(A) 1

(B) 2

(C) 3

(D) 4

Q.32

Calculate the quantity of CO_2 required to prepare $1\ L$ of soda water when the soda water was packed under $2\ atm$ of CO_2 . [Henry's law constant for CO_2 is $1.67 \times 10^8\ Pa$]

(A) 5.98 g

(B) 1.21 g

(C) 2.9 g

(D) 67.1 g

Q.33

Which of the following statements are not correct?

(A) All $C-O$ bonds in CO_3^{2-} are equal but not in H_2CO_3

(B) All $C-O$ bonds in HCO_2^- are equal but not in HCO_2H

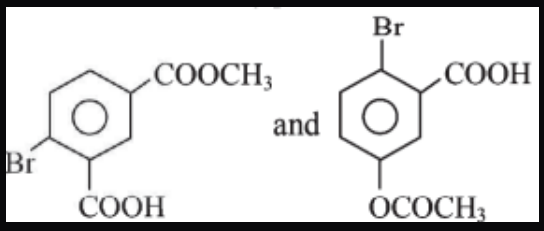
(C) $C-O$ bond length in HCO_2^- is longer than $C-O$ bond length in CO_3^{2-}

(D) None of these

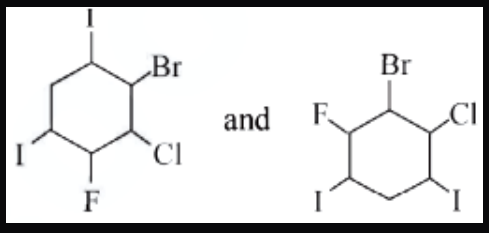
Q.34

Which of the following pair/s have same IUPAC naming :

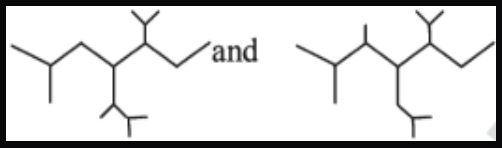
(A)



(B)



(C)



(D) None of these

Q.35

Which of the following lanthanoid ions is diamagnetic?
(At nos. $Ce = 58, Sm = 62, Eu = 63, Yb = 70$)

(A) Sm^{2+}

(B) Eu^{2+}

(C) Yb^{2+}

(D) Ce^{2+}

Q.36

Consider the following complexes.
(i) $[FeIF(CN)(H_2O)(en)]$
(ii) $[MoCl_2F_2(gly)]^2$
Then, calculate value of $|x-y|$ (where x and y are total number of possible optically active isomers in (i) and (ii) complex respectively).

(A) 15

(B) 12

(C) 8

(D) 6

Q.37

Oxygen is more electronegative than sulphur. Yet H_2S is acidic while H_2O is neutral. This is because

(A) Water is a highly associated compound

(B) Molecular mass of H_2S is more than that of H_2O

(C) H_2S is gaseous under ordinary conditions while H_2O is a liquid

(D) $H-S$ bond is weaker than $H-O$ bond

Q.38

N-Ethyl pthalimide on hydrolysis gives-

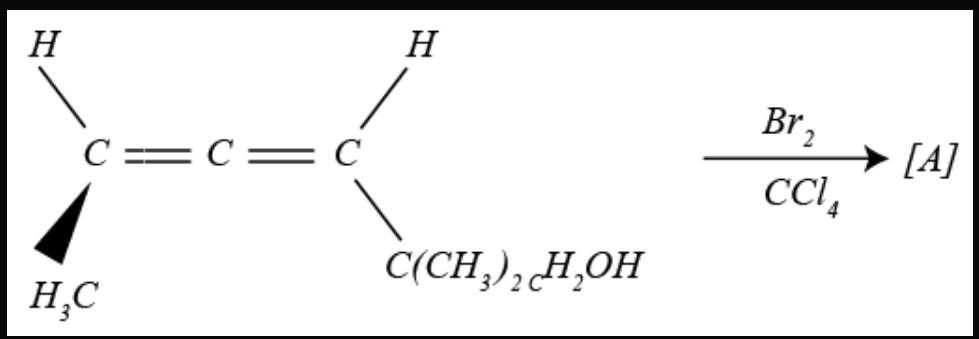
(A) Methyl alcohol

(B) Ethyl amine

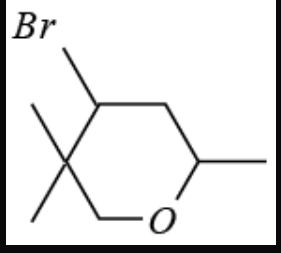
(C) Dimethyl amine

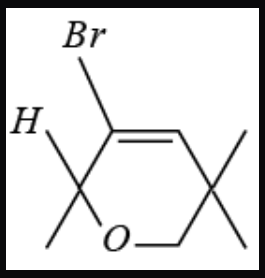
(D) Diethyl amine

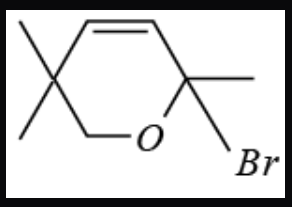
Q.39

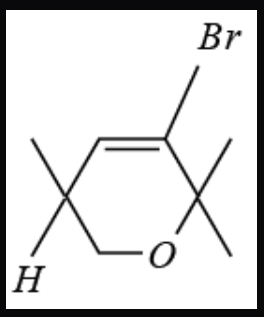


; Product $[A]$ of the reaction is:

(A) 

(B) 

(C) 

(D) 

Q.40

In the electrolysis of $CuSO_4$ solution, how many grams of Cu are plated out on the cathode in the time that it takes to liberate 5.6 litre of $O_2(g)$, measured at NTP at the anode?

(A) 31.75

(B) 14.2

(C) 4.32

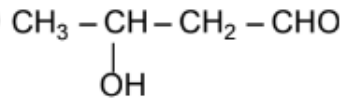
(D) None of these

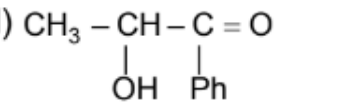
Q.41

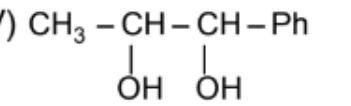
Identify the option which represents the correct products of the following reaction,

$Ph - CH = O + CH_3CH = O \xrightarrow{DilNaOH}$ (Aldols)

(I) 

(II) 

(III) 

(IV) 

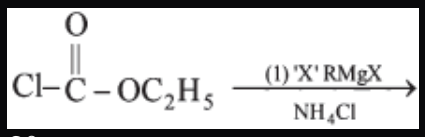
(A) I, II only

(B) I, II, III only

(C) III, IV only

(D) I, III only

Q.42



3° Alcohol.

Find out value of 'X'.

(A) 2

(B) 4

(C) 3

(D) 1

Q.43

The auto catalyst in the redox reaction involving acidified potassium permanganate and oxalic acid is

(A) Mn^{2+}

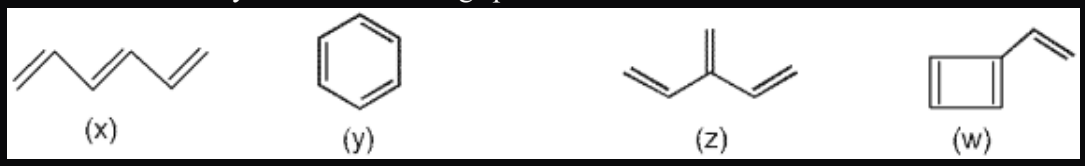
(B) SO_4^{2-}

(C) CO_2

(D) H_2O

Q.44

The correct stability order of following species is



(A) $x > y > w > z$

(B) $y > x > z > w$

(C) $x > w > z > y$

(D) $z > x > y > w$

Q.45

Match the following

Column-I Complex		Column-II Structure / Geometry / Property
(A) $[Ni(CN)_4]^{2-}$	(I)	Tetrahedral / Paramagnetic
(B) $[Ni(CO)_4]$	(II)	Tetrahedral / Diamagnetic
(C) $[NiCl_4]^{2-}$	(III)	Square planar / Diamagnetic

The correct match is

(A) A B C
II I III

(B) A B C
I. II III

(C) .A B C
III II I

(D) A B C
III I II.

Q.46

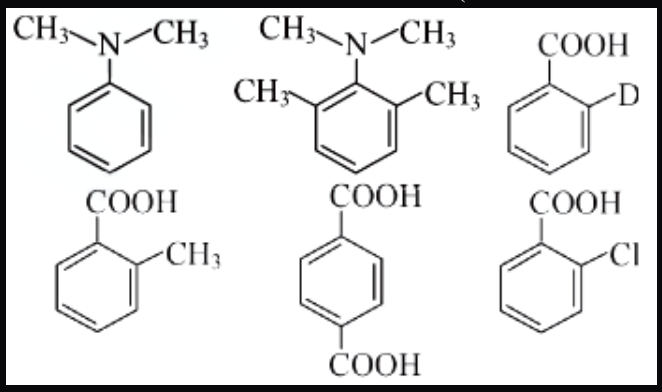
Barium ions, CN^- and Co^{2*} form an ionic complex. If that is supposed to be 75% ionised in water with vant Hoff factor (i) equal to four, then the coordination numbers of Co^{2+} in the complex can be :

Q.47

The number of complementary hydrogen bond(s) between a guanine and cytosine pair is

Q.48

The no. of structure those shows SIR (Steric inhibition of resonance)

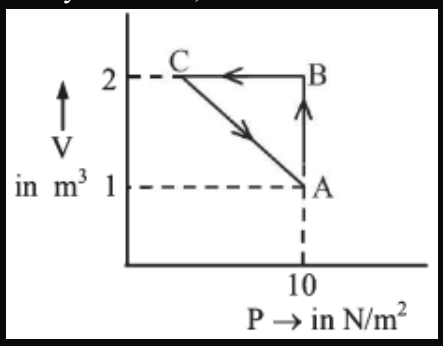


Q.49

In the Zerewitnoff's determination of active hydrogens in an alcohol $(\mathrm{Z}), 0.225\ \mathrm{g}$ of compound Z was treated with $\mathrm{CH_3MgBr}$ when $224\ \mathrm{mL}$ of $\mathrm{CH_4}$ was evolved at STP. A rest determination showed that Z had molecular weight of 180. How many acidic hydrogens are present per molecule of the compound Z ?

Q.50

An ideal gas is taken through the cycle $\mathrm{A} \longrightarrow \mathrm{B} \longrightarrow \mathrm{C} \longrightarrow \mathrm{A}$ as shown in the figure. If net heat supplied to the gas in the cycle is $5\ \mathrm{J}$, the work done on the gas in the process $\mathrm{C} \rightarrow \mathrm{A}$ is $\mathrm{J}-$



Q.51

The coefficient of t^8 in the expansion of $\left(1+2t^2-t^3\right)^9$, is

(A) 1680

(B) 2140

(C) 2520

(D) 2730

Q.52

What is the sum of all two digit numbers which give a remainder of 4 when divided by 6 ?

(A) 777

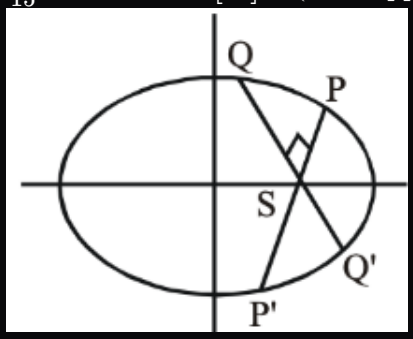
(B) 776

(C) 780

(D) 784

Q.53

In an ellipse $\frac{x^2}{10}+\frac{y^2}{6}=1$. If focal chords PSP' and QSQ' are at right angles to each other, then $\frac{1-e^2}{(SP)(SP')}+\frac{1-e^2}{(SQ)(SQ')}=\frac{N}{15}$. The value of $[N]$ is (where [.] is greatest integer function)



(A) 2

(B) 4

(C) 3

(D) 1

Q.54

Let the line $y=\log(a)(a>0,a\neq 1)$ intersects the line $y=x$ at point P. If point P lies on parabola $y^2=x$, then focal distance of point P is -

(A) e

(B) $e+\frac{1}{4}$

(C) 1

(D) $\frac{5}{4}$

Q.55

If α,β,γ are roots of the equation $x^3+x^2+2x-4=0$, then value of $\begin{vmatrix}1+\alpha^2 & 1 & 1\\ 1 & 4-\beta^3 & 1\\ 1 & 1 & 4-\gamma^3\end{vmatrix}$ (where α is real) -

(A) 0

(B) 40

(C) 10

(D) -15

Q.56

A variable line $\frac{x}{a}+\frac{y}{b}=1$ slides on the coordinate axes in second quadrant such that sum of length of intercepts made by it on axes is ℓ, then the locus of circumcentre of triangle made by the line with the axes is -

(A) $2(x-y)+\ell=0$

(B) $2(x+y)-\ell=0$

(C) $2(x+y)+\ell=0$

(D) None of these

Q.57

If (x, y) is a variable point on the curve $x^2 + y^2 - 2x - 2y - 2 = 0$, then minimum value of the expression $\frac{8}{(x-1)^2} + \frac{(y-1)^2}{4}$ is equal to -

(A) -2

(B) -1

(C) 1

(D) 2

Q.58

Let $f(\mathrm{x})$ be a twice differentiable function satisfying $\mathrm{x}f\left(\mathrm{x}^2\right) + \left(\mathrm{x}^2 + 1\right) f(\mathrm{x}) = \mathrm{x}$, then the value of $|9f''(1)|$ is

(A) 6

(B) 9

(C) 3

(D) 1

Q.59

Consider three vectors $\overrightarrow{\mathrm{a}}, \overrightarrow{\mathrm{b}}, \overrightarrow{\mathrm{c}}$ such that $|\overrightarrow{\mathrm{a}}| = 2, |\overrightarrow{\mathrm{b}}| = 1, |\overrightarrow{\mathrm{c}}| = 4$ angle between $\overrightarrow{\mathrm{a}} \& \overrightarrow{\mathrm{b}}$ is 120° and $\overrightarrow{\mathrm{c}}$ lies along the bisector of $\vec{a} \& \vec{b}$. Which of the following is true?

(A) $2\vec{a} - \vec{c} + 4\vec{b} = 0$

(B) $\vec{a} - 3\vec{c} - 4\vec{b} = 0$

(C) $3\vec{a} - \vec{c} - 4\vec{b} = 0$

(D) $\vec{a} + \vec{b} + \vec{c} = 0$

Q.60

Let f, g and h are differentiable function such that $g(x) = f(x) - x$ and $h(x) = f(x) - x^3$ are both strictly increasing functions, then the function $F(x) = f(x) - \frac{\sqrt{3}x^2}{2}$ is

(A) Strictly increasing $\forall x \in R$

(B) Strictly decreasing $\forall x \in R$

(C) Strictly decreasing on $\left(-\infty, \frac{1}{\sqrt{3}}\right)$ and strictly increasing on $\left(\frac{1}{\sqrt{3}}, \infty\right)$

(D) Strictly increasing on $\left(-\infty, \frac{1}{\sqrt{3}}\right)$ and strictly decreasing on $\left(\frac{1}{\sqrt{3}}, \infty\right)$

Q.61

Function $f(\mathrm{x}) = \begin{cases} \operatorname{sgn}([\mathrm{x}]) \; ; & \mathrm{x} \neq \mathrm{I} \\ [\operatorname{sgn}(\mathrm{x})] & ; \mathrm{x} = \mathrm{I} \end{cases}$

is {where $\operatorname{sgn}(\,)$ denotes signum function & [.] denotes greatest integer function }

(A) Continuous but not differentiable at $x = 2$

(B) Discontinuous at every integer

(C) Non differentiable at $x = 0$ & 1 only

(D) Non differentiable at every natural number

Q.62

If $f(\theta) = \frac{1-\sin 2\theta+\cos 2\theta}{2\cos 2\theta}$ then value of $f\left(11^\circ\right) \cdot f\left(34^\circ\right)$ equals

(A) $\frac{1}{2}$

(B) $\frac{3}{4}$

(C) $\frac{1}{4}$

(D) 1

Q.63

Twenty persons arrive in a town having 3 hotels x, y and z. If each person randomly chooses one of these hotels, then what is the probability that atleast 2 of them goes in hotel x, atleast 1 in hotel y and atleast 1 in hotel z ? (each hotel has capacity for more than 20 guests)

(A) $\frac{{}^{18}\mathrm{C}_2}{{}^{22}\mathrm{C}_2}$

(B) $\frac{{}^{20}\mathrm{C}_2 . {}^{18}\mathrm{C}_1 . {}^{17}\mathrm{C}_1 . 3^{16}}{3^{20}}$

(C) $\frac{{}^{20}\mathrm{C}_9}{3^9}$

(D) $\frac{3^{20}-13.2^{20}+43}{3^{20}}$

Q.64

The value of the integral $\int_{\frac{-(\pi-2)}{4}}^{\frac{\pi-1}{2}} \left[\sin^{-1}(\sin(2x+1))\right] dx$ (where [.] denotes greatest integer function) is equal to

(A) $\pi - 2$

(B) 2

(C) $\frac{\pi}{2} - 1$

(D) 1

Q.65

There is a complex number z with imaginary part 164 and a positive integer n such that $\frac{\mathrm{z}}{\mathrm{z}+\mathrm{n}} = 4i$. The value of n is equal to

(A) 615

(B) 707

(C) 697

(D) 867

Q.66

Let $f(x)=\begin{cases} \mathrm{x}, \mathrm{x}<0 \\ 1+\mathrm{x}^2, \mathrm{x} \geq 0 \end{cases}$ and $\mathrm{g}(\mathrm{x})=1+\mathrm{x}-[\mathrm{x}]$, then range of $f \circ g(x)$ is (where [.] denotes greatest integer function)-

(A) $[1,5]$

(B) $[2,5)$

(C) $(2,5]$

(D) $[1,5)$

Q.67

$A_1=[a_1]$

$A_2=\begin{bmatrix} a_2 & a_3 \\ a_4 & a_5 \end{bmatrix}$

$A_3=\begin{bmatrix} a_6 & a_7 & a_8 \\ a_9 & a_{10} & a_{11} \\ a_{12} & a_{13} & a_{14} \end{bmatrix} \ldots\ldots\ldots\ldots\ldots\ldots\ldots A_n=[\ldots\ldots\ldots]$

Where $a_r=[\log_2 r]$ ([.] denotes greatest integer). Then trace of A_{10} is equal to

(A) 100

(B) 80

(C) 120

(D) 150

Q.68

Let $f(x)=\begin{bmatrix} x^2+a & ,0 \leq x<1 \\ 2x+b & ,1 \leq x \leq 2 \end{bmatrix}$ and $g(x)=\begin{bmatrix} 3x+b, & 0 \leq x<1 \\ x^3 & ,1 \leq x \leq 2 \end{bmatrix}$
If derivative of $f(x)$ w.r.t. $g(x)$ at $x=1$ exists, and has value equal to m then the value of $|a+b+6m|$ is equal to

(A) 1

(B) 2

(C) 1.5

(D) 0.33

Q.69

If $\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{x^4}{1-x^4} \cos^{-1} \frac{2x}{1+x^2} dx=k \int_0^{1/\sqrt{3}} \frac{x^4}{1-x^4} dx$ then ' k ' equals

(A) π

(B) 2π

(C) 2

(D) 1

Q.70

If the foot of the perpendicular from the point $A(p+1,-1,11)$ on the line $\frac{x}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ is $\mathrm{B}(\mathrm{q}, 5,7)$ then the value of $(\mathrm{p}-\mathrm{q})$ is ....

(A) 1.5

(B) 1

(C) 0.5

(D) 0

Q.71

If Mr. Kohli is moving inside the area bounded by the three curves denoted by $x^2y^2+x^4-3x^2-y^2+2=0$ is $\pi+k$, then the value of k is

Q.72

Let $\mathrm{x}_1, \mathrm{x}_2, \ldots \ldots \ldots, \mathrm{x}_{100}$ are 100 observations such that $\sum \mathrm{x}_{\mathrm{i}}=0, \sum_{1 \leq i<j \leq 100}\left|\mathrm{x}_{\mathrm{i}} \mathrm{x}_{\mathrm{j}}\right|=80000$ \& mean deviation from their mean is 5 , then their standard deviation, is-

Q.73

If the sum of the coefficients in the expansion of $(1+2x)^n$ is 6561 and the greatest term in the expansion, for $x=\frac{1}{2}$, is the kth term, then k^k is equal to

Q.74

If $\frac{x^2+y^2}{x^2-y^2}+\frac{x^2-y^2}{x^2+y^2}=2$ then find value of $\frac{x^8+y^8}{x^8-y^8}+\frac{x^8-y^8}{x^8+y^8}$

Q.75

If $f(x)=\mathrm{e}^{(\mathrm{x}+1)^{\mathrm{n}}};(\mathrm{n} \in \mathrm{N})$, then value of ' n ' for which $f''(1)=67\left(2^n \mathrm{e}^{2^n}\right)$ is-

Answers & Solutions

Q.1 Answer:

greater than $v\sqrt{\frac{4}{3}}$

Solution:

$$\frac{1}{2}mv^2=\frac{hc}{\lambda}-\phi$$

$$\frac{1}{2}mv'^2=\frac{hc}{(3\lambda/4)}-\phi=\frac{4hc}{3\lambda}-\phi$$

Clearly $v'>\sqrt{\frac{4}{3}}v$

Q.2 Answer:

NAND gate

Solution:

The truth table for the waveform is given by

Misplaced \hline

The expression for the output is $Y = \overline{A \cdot B}$

The truth table of the given logic gate matches with NAND gate.

Q.3 Answer:

100

Solution:

We have $K_\alpha = \frac{m_y}{m_y + m_\alpha} \cdot Q$

$\Rightarrow K_\alpha = \frac{A-4}{A} \cdot Q$

$\Rightarrow 48 = \frac{A-4}{A} \cdot 50$

$\Rightarrow A = 100$

Q.4 Answer:

$\frac{I_0}{4}$

Solution:

For a Young's double slit experiment....

$\beta = \frac{\lambda D}{d}$

$y = \frac{\beta}{3} = \frac{\lambda \mathrm{D}}{3\ \mathrm{d}}$

$\phi = 2\pi/3$

$\mathrm{I} = \mathrm{I}_0 \cos^2(\phi/2)$

$= \mathrm{I}_0 \cos^2(\pi/3) = \mathrm{I}_0/4$

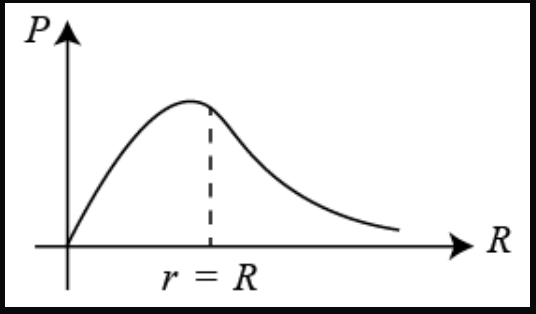
Q.5 Answer:

power consumed by it will decrease

Solution:

Power maximum when $r = R$.

So, power consumed by it will decrease. for $R > r$.



Q.6 Answer:

spring will be in its natural length under all conditions

Solution:

Let the value of ' a ' be increased from zero.

As long as a $\leq \mu g$, there shall be no relative motion between m_1 or m_2 and platform, that is, m_1 and m_2 shall move with acceleration a.

As $a > \mu g$ the acceleration of m_1 and m_2 shall become μg each. Hence at all instants the velocity of m_1 and m_2 shall be same

$\therefore$ The spring shall always remain in natural length.

Q.7 Answer:

2

Solution:

The escape velocity for the surface of earth is or

$\mathrm{v_{se(e)}} = \sqrt{\frac{2\mathrm{GM_e}}{\mathrm{R_e}}}$

or $\mathrm{v_{es(e)}} = \mathrm{R_e}\sqrt{\frac{8}{3}\mathrm{G}\pi\rho} \quad \left[\because \mathrm{M_e} = \frac{4}{3}\pi\mathrm{R}^3\right]$

$-\mathrm{Re_e}\sqrt{\frac{8}{3}\mathrm{G}\pi\rho}$

$\frac{v_{es(e)}}{v_{es(P)}} = \frac{R_e\sqrt{\frac{8}{3}G\pi\rho}}{4R_e\sqrt{\frac{8}{3}G\pi 9\rho}} = \frac{1}{12}$

So, $\mathrm{v_{es(p)}} = 12\mathrm{v_{es(e)}}$

Q.8 Answer:

increases 32 times

Solution:

$\because B = \frac{\mu_0 I}{2r} \quad \text{and} \quad I = \frac{e}{T}$

$B = \frac{\mu_0 e}{2rT} \quad \left[r \propto n^2, T \propto n^3\right]$

$\Rightarrow B \propto \frac{1}{n^5}$

Q.9 Answer:

2.5 A

Solution:

$P = VI$

For secondary :

$V_2 = \frac{P_2}{I_2} = \frac{500}{12.5} = 40 \text{ volts}$

For an ideal transformer (100% efficient)

$P_{\text{input}} = P_{\text{output}}$

$\Rightarrow \quad V_1 I_1 = V_2 I_2$

$\Rightarrow I_1 = \frac{V_2 I_2}{V_1} = \frac{40(12.5)}{40 \times 5} = 2.5 \text{ A} \quad \left[\because \frac{n_1}{n_2} = \frac{V_1}{V_2} \quad \Rightarrow \frac{5}{1} = \frac{V_1}{40}\right].$

Q.10 Answer:

$\frac{2Q}{3}$

Solution:

$T_A = T_C$

$\Delta U_{BC} = -Q$

$\Delta U_{AB} = Q$

$\Rightarrow n\frac{3}{2}R\Delta T = Q$

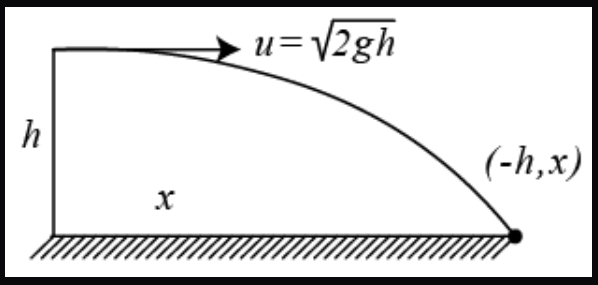
$W_{AB} = nR\Delta T = 2\frac{Q}{3}$

Q.11 Answer:

$2\ h$

Solution:

Using equation of trajectory:



$-h = x\tan(0^\circ) - \frac{gx^2}{2(2gh)(\cos^2 0^\circ)} \quad \Rightarrow x = 2h$

Method II

time of flight $T = \sqrt{\frac{2h}{g}}$ **horizontal distance covered during time of flight is**

$x = u_x t = \sqrt{\frac{2h}{g}} \times \sqrt{2hg} = 2h$

Q.12 Answer:

$[M^0\ L^2\ T^0]$

Solution:

$\left[\frac{\alpha z}{k\theta}\right] = [M^0 L^\circ T^\circ] \quad \Rightarrow \quad [\alpha] = \left[\frac{k\theta}{z}\right]$

$\text{Further } [P] = \left[\frac{\alpha}{\beta}\right] \quad \Rightarrow \quad \therefore \quad [\beta] = \left[\frac{\alpha}{P}\right] = \left[\frac{k\theta}{ZP}\right]$

Dimensions of $K\theta$ **is that to energy. Hence,**

$[\beta] = \left[\frac{ML^2T^{-2}}{LML^{-1}T^{-2}}\right] = [M^0L^2T^0]$

Q.13 Answer:

100% increase

Solution:

We have $\rho = \frac{PM}{RT}$

$\frac{P_1 M}{RT_1} = \frac{P_2 M}{RT_2}$

$\frac{P_1}{T_1} = \frac{P_2}{2T_1}$

$P_2 = 2P_1$

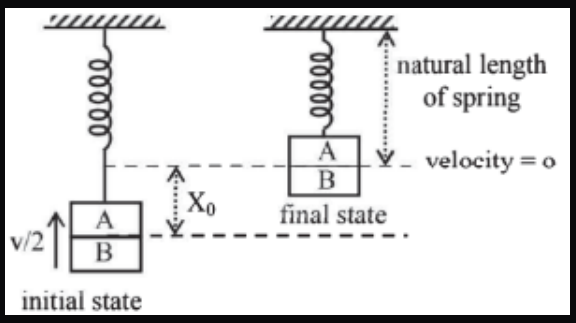
i.e., change in pressure is 100%**.**

Q.14 Answer:

$\sqrt{\frac{6mg^2}{k}}$

Solution:

The initial extension in spring is $X_0 = \frac{mg}{k}$. Just after collision of B with A the speed of combined mass is $\frac{v}{2}$.
For the spring to just attain natural length the combined mass must rise up by $X_0 = \frac{mg}{k}$ and comes to rest.



Applying conservation of energy between initial and final states

$\frac{1}{2}2\ m\left[\frac{v}{2}\right]^2 + \frac{1}{2}k\left[\frac{mg}{k}\right]^2 = 2mg\left[\frac{mg}{k}\right]$

Solving we get $v = \sqrt{\frac{6mg^2}{k}}$.

Q.15 Answer:

4

Solution:

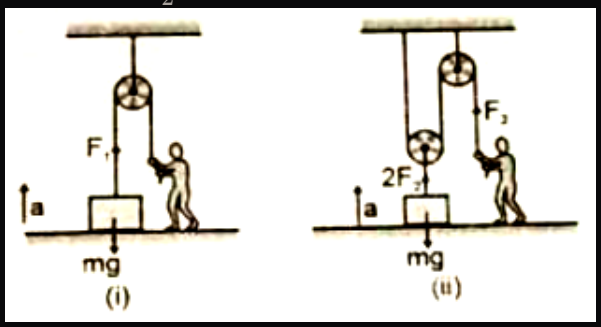
The digits $4, 1, 5$, and 0 are significant because they convey meaningful information about the value.
The exponent, -4, does not count as a significant figure because it is not a digit that represents the precision of the measurement; it indicates the order of magnitude (in scientific notation).
So, there are four significant figures in the number 4.150×10^{-4}.

Q.16 Answer:

(i)

Solution:

Since, $h = \frac{1}{2}at^2 \Rightarrow a$ should be same in both cases, because h and t are same in both cases as given.



In (i) $F_1 - mg = ma. \quad \Rightarrow \quad F_1 = mg + ma$
In (ii) $2F_2 - mg = ma \quad \Rightarrow \quad F_2 = \frac{mg+ma}{2} \quad \therefore \quad F_1 > F_2.$

Q.17 Answer:

Statement-1 is False, Statement-2 is True.

Solution:

Equivalent thermal conductivity of two identical rods in series is given by

$\frac{2}{\mathrm{K}}=\frac{1}{\mathrm{K}_1}+\frac{1}{\mathrm{K}_2}$

If $K_1<K_2$, then $K_1<K<K_2$

Hence statement 1 is false.

Q.18 Answer:

$\frac{-qQ}{6\pi\varepsilon_0\,\mathrm{L}}$ and zero

Solution:

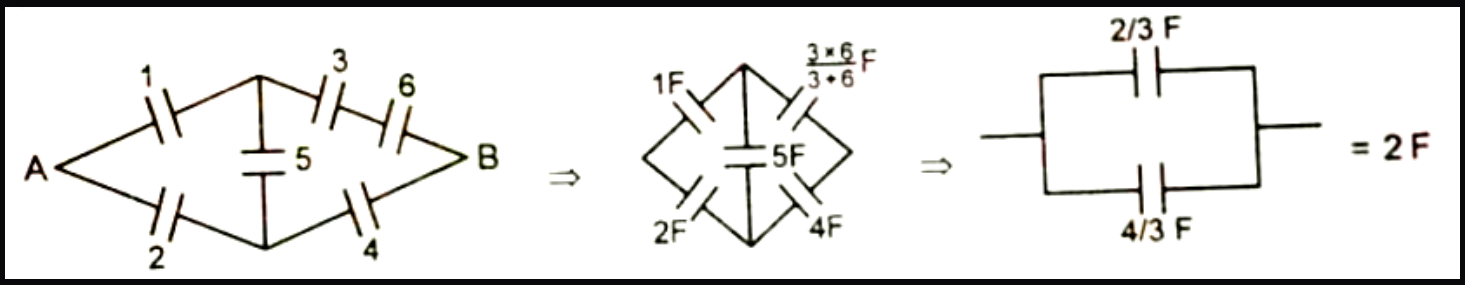
$$\mathrm{W_{CRD}}=\mathrm{Q}\left(\mathrm{V_D}-\mathrm{V_C}\right)$$
$$=\mathrm{Q}\left[\mathrm{V_D}-0\right]$$
$$(\because \text{ potential at centre of dipole is zero })$$
$$=\left[\frac{\mathrm{kq}}{3\,\mathrm{L}}-\frac{\mathrm{kq}}{\mathrm{L}}\right]\mathrm{Q}$$
$$\mathrm{W_{CRD}}=-\frac{\mathrm{kqQ}}{\mathrm{L}}\times\frac{2}{3}=-\frac{\mathrm{qQ}}{6\pi\varepsilon_0\,\mathrm{L}}$$

Q.19 Answer:

2 F

Solution:

Equivalent circuit is



Q.20 Answer:

$2\pi\sqrt{\frac{\ell}{g}}$

Solution:

After first collision B reaches up to angular displacement α and after second collision ' B ' reaches to it's initial position.

So, total time taken is $\mathrm{t}=\frac{T}{4}+\frac{T}{4}+\frac{T}{4}+\frac{T}{4}=\mathrm{T}$

Q.21 Answer:

—

Solution:

$f\propto\frac{1}{\ell}$

Let the freq. of tuning fork be f

Earlier the freq. of the string is $f-4$ & later it is equal to f

$$\therefore \quad \frac{f-4}{f}=\frac{99}{100}$$

$$\Rightarrow \quad 99f=100f-400 \quad \Rightarrow \quad f=400$$

Q.22 Answer:

—

Solution:

$$\mathrm{I}=\frac{2}{5}\,\mathrm{m}\left(\frac{\mathrm{R}}{2}\right)^2+\mathrm{m}(2\mathrm{R})^2+\frac{2}{3}\,\mathrm{m}\left(\frac{\mathrm{R}}{2}\right)^2$$

$$=\frac{21}{5}\mathrm{mR}^2$$

Q.23 Answer:

—

Solution:

Due to error in

$$\ell\left(\frac{\Delta\rho}{\rho}\right)_{\max}=\frac{\Delta\ell}{\ell}+\frac{\Delta\ell}{100-\ell}$$

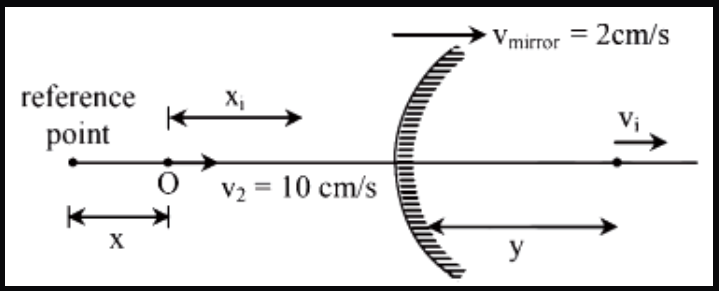
$$=\frac{\Delta\ell(100)}{\ell(100-\ell)}$$

When $\ell(100-\ell)$ is maximum then $\left(\frac{\Delta\rho}{\rho}\right)_{\max}$ will be minimum, that means $\ell=50\ \mathrm{cm}$

Q.24 Answer:

—

Solution:



$$\frac{1}{y}+\frac{1}{-x_1}=\frac{1}{10}\ \text{(i)}$$

$$\frac{1}{y}-\frac{1}{10}=\frac{1}{10}\Rightarrow y=5\ \mathrm{cm}$$

$$\frac{dx}{dt}=10\ \mathrm{cm/s},\frac{d\left(x+x_1\right)}{dt}=2\ \mathrm{cm/s}$$

$$\Rightarrow \frac{dx_1}{dt}=-8\ \mathrm{cm/s}$$

$$\frac{dy}{dt}=v_1-2$$

From (i), $\frac{dy}{dt} = \left(\frac{y}{x_1}\right)^2 \frac{dx_1}{dt}$

$\Rightarrow v_1 - 2 = \left(\frac{5}{10}\right)^2 (-8) \Rightarrow v_1 = 0$

Q.25 Answer:

—

Solution:

Since current $I = neAv_d$ through both rods is same $2(n)$ eAv $v_L = n \in (2A)v_R$

$\Rightarrow$ or $\frac{V_L}{V_R} = 1$

Q.26 Answer:

$S^{2-} > K^+ > Sc^{3+} > V^{5+} > Mn^{7+}$

Solution:

Ionic radius is inversely proportional to charge present on metal respectively.

In transition metal series, ionic radius will decreases gradually fromSc to Mn and constant in middle fromFe to Co and again increase from Ni to

Zn.

Hence, $Sc^{3+} > V^{5+} > Mn^{7+}$

Greater the oxidation state, lesser will be the ionic radii.

S^{2-} acquire very high charge in it. It increases our ionic radii than K^+. Therefore, S^{2-} is very big in size.

Hence, correct order of ionic radii is

$S^{2-} > K^+ > Sc^{3+} > V^{5+} > Mn^{7+}$

Q.27 Answer:

ionic, covalent, coordinate covalent and hydrogen

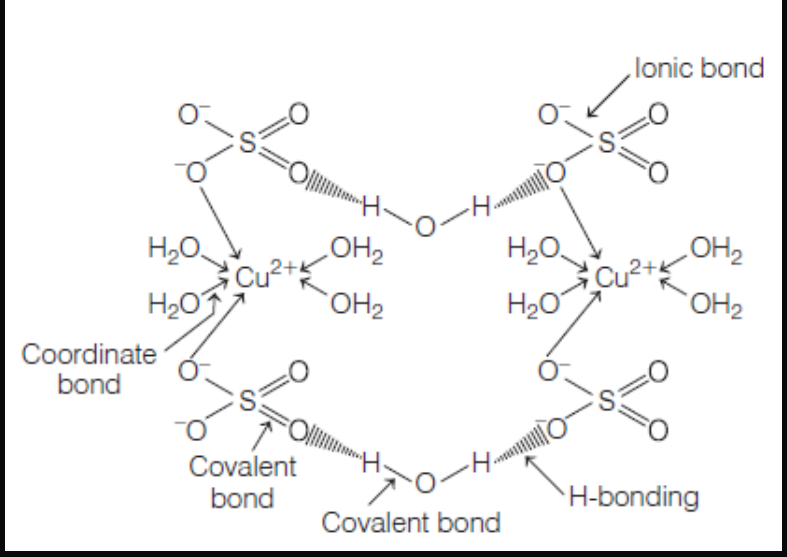
Solution:

According to structure of $CuSO_4 \cdot 5H_2O$. A covalent bond is present between sulphur and oxygen as they are non-metals and the bond between copper and sulphate is ionic i.e. one is

metal and other is non-metal.

Hydrated copper sulphate is a complex which is bonded to water molecules acting as ligand forming coordinate bond.

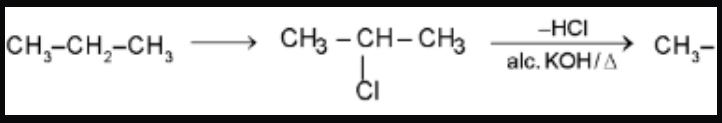
The Cu^{2+} ions from coordinate covalents bonds with water as well as sulphate ions.



Q.28 Answer:

$Cl_2/h\nu$, alc. $KOH/\Delta, HBr/R_2O_2$

Solution:

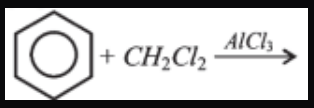


$$CH = CH_2 \xrightarrow{HBr, R_2O_2(AMK)} CH_3 - CH_2 - CH_2 - Br.$$

Q.29 Answer:

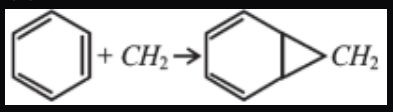
All of these

Solution:

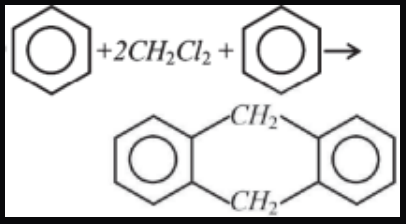


Many products are there

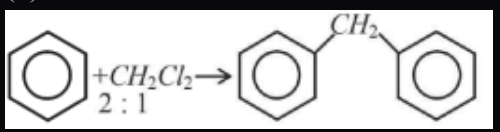
(1)



(2)



(3)



All of these

Q.30 Answer:

5

Solution:

$X_A = \frac{P_A}{P_T}$ or $X_B = \frac{P_B}{P_T}(1 - X_A) = \frac{P_B}{P_T}$

$\frac{P_A}{P_B} = \frac{X_A}{(1 - X_A)}$

For liquid phase $\mathrm{P_A = P_A^\circ X_A}$

$P_B = P_B^\circ X_B = P_B^\circ (1 - X_A)$

$\frac{P_A}{P_B} - \frac{60 - 0.2}{15 - (1 - 0.2)} - 1$

$\frac{X_A}{1 - X_A} = 0.5$ compare with 0.2

$X = 5$

Q.31 Answer:

1

Solution:

1. Shielding effect is effective, when orbitals in the inner shells are completely filled. So, statement (1) is correct.

2. Inner shells are show effect. So, statement 2 is incorrect.

3. Shielding effect as well as ionisation energy decrease down the group. So, statement 3 is incorrect.

4. With increase in nuclear charge, shielding effect decreases. So, statement 4 is incorrect.

Q.32 Answer:

2.9 g

Solution:

According to Henry's law;

$\mathrm{x = \frac{P}{K_H} = \frac{2 \times 1.013 \times 10^5}{1.67 \times 10^8}}$

$\frac{\mathrm{n_{CO_2}}}{1000/18} = \frac{2 \times 1.013}{1.67 \times 1000}$

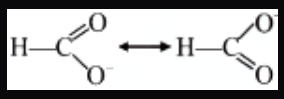
$\mathrm{n_{CO_2}} = \frac{2 \times 1.013}{1.67 \times 18} \Rightarrow$ moss of CO_2 dissolved

$= \frac{2 \times 1.013 \times 44}{1.67 \times 18} = 2.9\ \mathrm{g}$

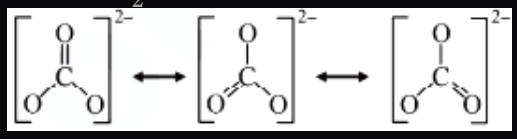
Q.33 Answer:

$\mathrm{C - O}$ bond length in $\mathrm{HCO_2^-}$ is longer than $\mathrm{C - O}$ bond length in $\mathrm{CO_3^{2-}}$

Solution:



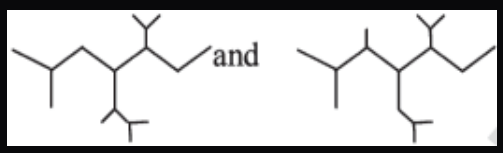
B.O. $= \frac{2+1}{2} = 1.5$



$\text{B.O.} = \frac{2+1+1}{3} = \frac{4}{3} = 1.33$

$\mathrm{C-O}$ **bond length in** $\mathrm{HCOO^-}$ **is less than** $\mathrm{C-O}$ **bond length in** $\mathrm{CO_3^{2-}}$

Q.34 Answer:



Solution:

[3] Both have same IUPAC name

3-ethyl-2,5,6-trimethyl-4-[methyl propyl] heptane

[4] Both have same IUPAC name

4-amino-2-chloro methl-3-fluoromethyl butan-1-ol

Q.35 Answer:

$\mathrm{Yb^{2+}}$

Solution:

$\mathrm{Sm^{+2}}(Z=62) \quad [\mathrm{Xe}]4\mathrm{f}^6$

$\mathrm{Eu^{+2}}(Z=63) \quad [\mathrm{Xe}]4\mathrm{f}^7$

$\mathrm{Yb^{+2}}(Z=70) \quad [\mathrm{Xe}]4\mathrm{f}^{14}$

$\mathrm{Ce^{+2}}(Z=58) \quad [\mathrm{Xe}]4\mathrm{f}^1, 5\,\mathrm{d}'$

Only $\mathrm{Yb^{+2}}$ **is diamagnetic.**

Q.36 Answer:

8

Solution:

(i) $[\mathrm{FeIF(CN)\,(H_2O)\,(en)}]$

$\mathrm{x} = 12$ **(active isomer)**

(ii) $[\mathrm{MoCl_2\,F_2\,(gly)\,}]^2$

$\mathrm{y} = 4$ **(active isomer)**

$x - y = 12 - 4 \Rightarrow 8$

Q.37 Answer:

$\mathrm{H-S}$ bond is weaker than $\mathrm{H-O}$ bond

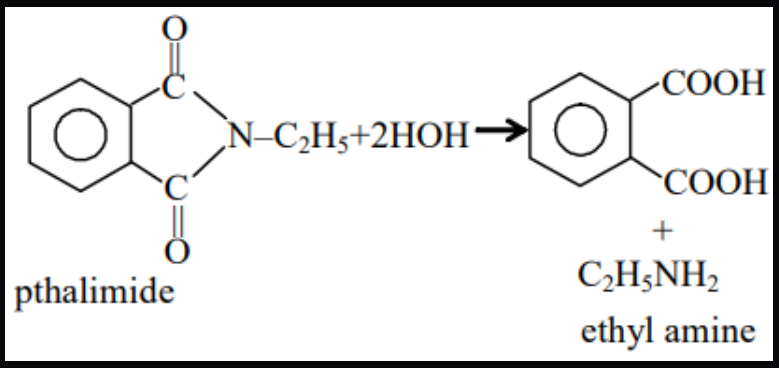
Solution:

H_2S is more acidic than H_2O. It is due to the fact than $H-S$ bond is weaker than $H-O$ bond.

Q.38 Answer:

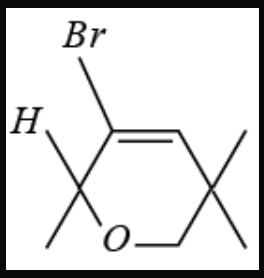
Ethyl amine

Solution:

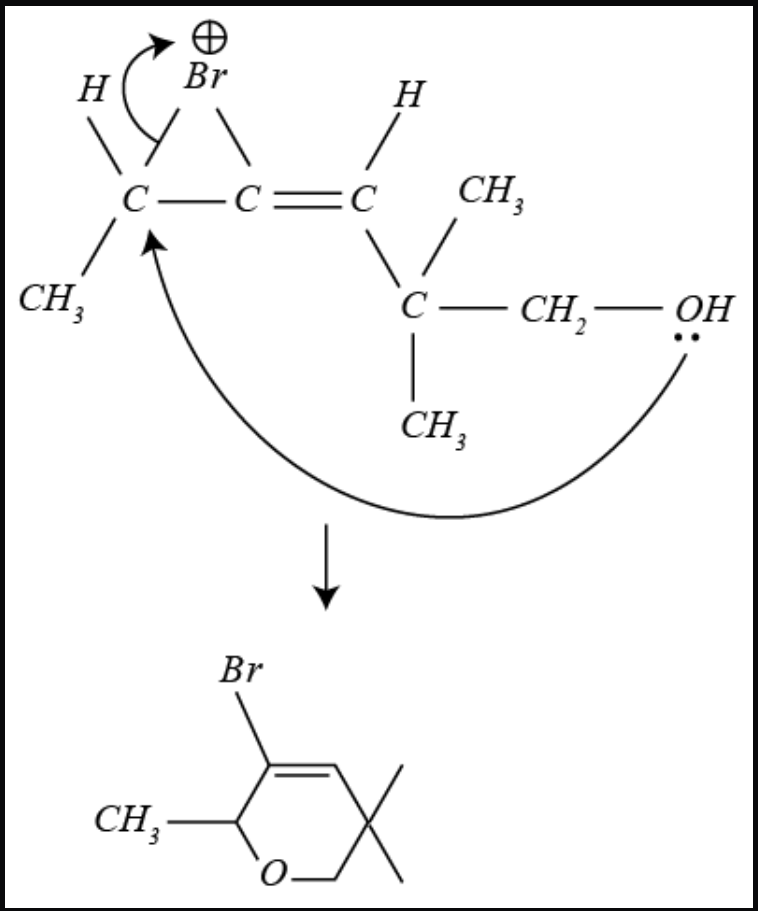


Therefore option (B) is correct.

Q.39 Answer:



Solution:



Q.40 Answer:

31.75

Solution:

At cathode: $\mathrm{Cu}^{+2}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$ **at anode :** $2\mathrm{H_2O}_{(l)} \longrightarrow \mathrm{O}_{2(g)} + 4\mathrm{H}^+{}_{(aq.)} + 4e^-$ **Number of equivalent of** Cu **= number of equivalent of** $\mathrm{O_2}$

$$n \times n_f)_{\mathrm{Cu}} = n \times n_f)_{\mathrm{O_2}}$$

$$n \times 2 = \frac{5.6}{22.4} \times 4$$

$$n = \frac{1}{2}$$

$$\therefore \text{ mass of Cu} = \frac{1}{2} \times 63.5\text{gm}$$

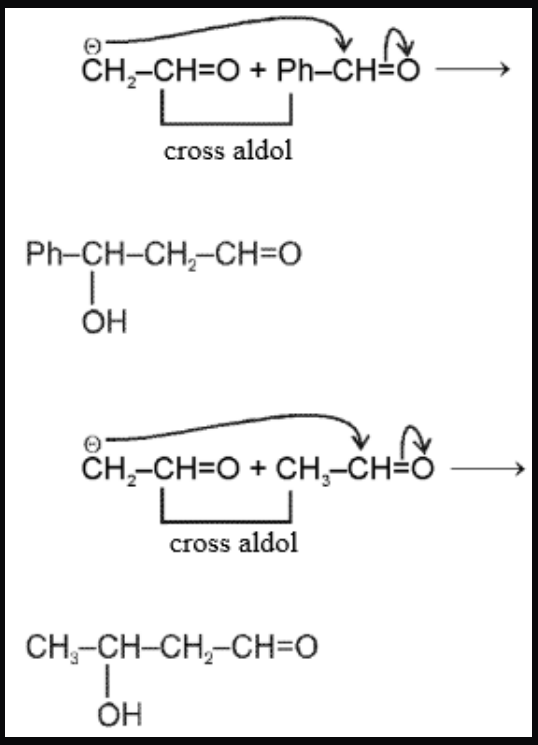
$$= 31.75\text{gm}$$

Q.41 Answer:

I, II only

Solution:

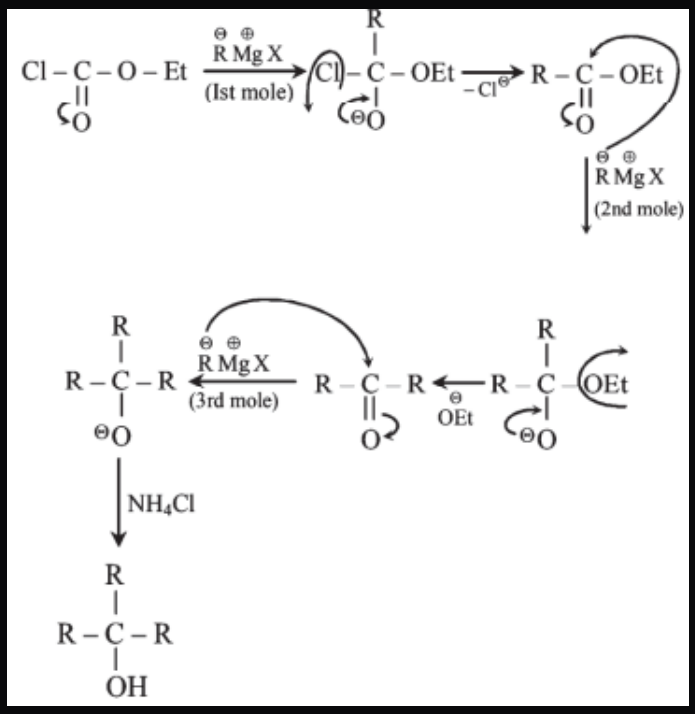
$$\mathrm{Ph - CH = O + CH_3CH = O} \xrightarrow{\mathrm{NaOH}}$$



Q.42 Answer:

3

Solution:



Q.43 Answer:

Mn^{2+}

Solution:

$5C_2O_4^{2-} + 2MnO_4^- + 16H^+ \rightarrow$

(Coloured)

$Mn^{2+} + 8H_2O + 10CO_2$

(Colourless)

Formation of Mn^{2+} ions, which is indicated by the decrease in the intensity of the colour of the solution, the rate of the reaction increases which was slow initially.

Q.44 Answer:

$y > x > z > w$

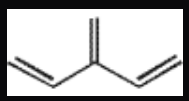
Solution:



$\rightarrow$ Aromatic compound



$\rightarrow$ Linear conjugation



$\rightarrow$ Cross conjugation



$\rightarrow$ Anti Aromatic

Q.45 Answer:

.A	B	C
III	II	I

Solution:

$[\mathrm{Ni}(\mathrm{CN})_4]^{2-}$ Square planar

The electrons are paired in $3\mathrm{d}$ orbitals because CN^- is strong field ligand.Hence the hybridisation will be dsp^2

$[\mathrm{Ni}(\mathrm{CO})_4]$ Tetrahedral

CO is strong field ligand. First the unpaired electrons will pair and then the electrons from $4\mathrm{s}$ orbital will go in $3\mathrm{d}$ orbital. Now $4\mathrm{s}$ and $4\mathrm{p}$ is empty hence it will be sp^3 hybridised.

$[\mathrm{NiCl}_4]^2$ is paramagnetic in nature due to presence of 2 unpaired electrons.

Ni^{2+} ion has $3\mathrm{d}^8$ electronic configuration. It undergoes sp^3 hybridisation which results in tetrahedral geometry.

Q.46 Answer:

—

Solution:

$$\alpha = \frac{\mathrm{i}-1}{\mathrm{x}-1} = \frac{3}{4} = \frac{4-1}{\mathrm{x}-1}$$

$\therefore \mathrm{x} = 5 \quad \therefore \quad \text{Complex is } \mathrm{Ba}_3\left[\mathrm{Co(CN)}_5\right]_2$

Q.47 Answer:

—

Solution:

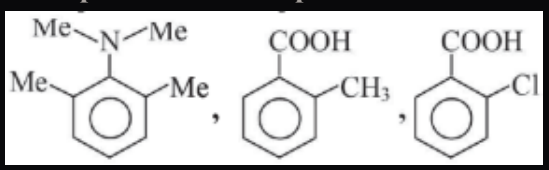
The number of hydrogen bonds between adenine and thymine are two $(A = T)$ where as the number of hydrogen bond between guanine and cytosine are three $(\mathrm{C} : \mathrm{C})$.

Q.48 Answer:

—

Solution:

Example where SIR is possible



Q.49 Answer:

—

Solution:

Let the compound Z be ROH

$\mathrm{ROH} + \mathrm{CH_3MgBr} \longrightarrow \mathrm{CH_4} + \mathrm{Mg(OH)Br}$

No. of mole of $Z = 0.00125$

No. of mole of $\mathrm{CH_4} = 0.01$

$\because 0.00125$ mol of alcohol gives 0.01 mol of $\mathrm{CH_4}$

$\therefore 1$ mol of alcohol gives 8 mol of $\mathrm{CH_4}$

Q.50 Answer:

—

Solution:

$\Delta W_{A \to B} = -10J$

$\Delta W_{AB} = P\Delta V; \Delta W_{BC} = 0$

$\Delta q = \Delta E - \Delta W$

$\Delta E = 0$

$\Delta q = 5 = 0 - \Delta W$

$\Delta W = -5J = \Delta W_{A \to B} + \Delta W_{B \to C} + \Delta W_{C \to A}$

or $-5J = -10 + 0 + \Delta W_{C \to A}$

$\Delta W_{C \to A} = 5J$

Q.51 Answer:

2520

Solution:

We have $\left((1+2t^2) - t^3\right)^9 = {}^9C_0\left(1+2t^2\right)^9 - {}^9C_1\left(1+2t^2\right)^8 \cdot t^3 + {}^9C_2\left(1+2t^2\right)^7 \cdot t^6 - {}^9C_3\left(1+2t$
${}^9C_9\left(t^3\right)^9$

$\therefore$ Coefficient of t^8 in the expansion of $\left(1+2t^2-t^3\right)^9$

$= {}^9C_0\left(\text{ coefficient of } t^8 \text{ in } \left(1+2t^2\right)^9\right) - {}^9C_1\left(\text{ coefficient of } t^5 \text{ in } \left(1+2t^2\right)^8\right) + {}^9C_2\left(\text{ coefficient }$

$= {}^9C_0 \cdot {}^9C_4 24 - 0 + {}^9C_2 \cdot {}^7C_1 \cdot 2 = \frac{9 \cdot 8 \cdot 7 \cdot 6}{4 \cdot 3 \cdot 2 \cdot 1} \cdot 16 + \frac{9 \cdot 8}{2 \cdot 1} \cdot 7 \cdot 2 = 9 \cdot 8 \cdot 7 \cdot 5 = 2520$ Ans.

Q.52 Answer:

780

Solution:

Such numbers are $10, 16, \ldots..94$

If n be the number of terms, then

$94 = \mathrm{t_n} = \mathrm{a} + (\mathrm{n}-1)\mathrm{d} = 10 + (\mathrm{n}-1)6 \Rightarrow \mathrm{n} = 15$

$\therefore \text{sum} = \frac{\mathrm{n}}{2}(\mathrm{a} + \ell) = \frac{15}{2}(10 + 94) = 780$

Q.53 Answer:

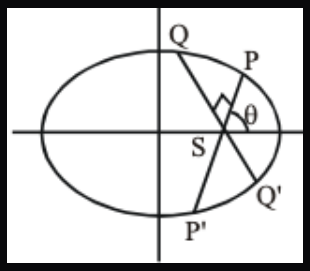
4

Solution:

Let PSP' is

$x = ae + r\cos\theta$

$y = r\sin\theta$



$$\frac{(\mathrm{ae} + \mathrm{r}\cos\theta)^2}{\mathrm{a}^2} + \frac{(\mathrm{r}\sin\theta)^2}{\mathrm{b}^2} = 1$$

$$\mathrm{r}^2\left(\frac{\cos^2\theta}{\mathrm{a}^2} + \frac{\sin^2\theta}{\mathrm{b}^2}\right) + \mathrm{r}(\lambda) + \frac{\mathrm{a}^2\mathrm{e}^2 - \mathrm{a}^2}{\mathrm{a}^2} = 0$$

$$|\mathrm{r}_1\mathrm{r}_2| = \frac{\left(1 - \mathrm{e}^2\right)\mathrm{a}^2\,\mathrm{b}^2}{\mathrm{b}^2\cos^2\theta + \mathrm{a}^2\sin^2\theta}$$

$$\frac{\mathrm{a}^2\sin^2\theta + \mathrm{b}^2\cos^2\theta}{\mathrm{a}^2\,\mathrm{b}^2} = \frac{1 - \mathrm{e}^2}{SP.SP'}$$

Replacing θ by $90 - \theta$.

$$\frac{\mathrm{b}^2\sin^2\theta + \mathrm{a}^2\cos^2\theta}{\mathrm{a}^2\,\mathrm{b}^2} = \frac{1 - \mathrm{e}^2}{\mathrm{SQ.SQ'}}$$

$$\frac{\mathrm{a}^2 + \mathrm{b}^2}{\mathrm{a}^2\,\mathrm{b}^2} = \frac{\mathrm{N}}{15}$$

$$\frac{16}{60} = \frac{\mathrm{N}}{15} \Rightarrow \mathrm{N} = 4$$

Q.54 Answer:

$\frac{5}{4}$

Solution:

$P \equiv (\log a, \log a)$

$\Rightarrow \log^2 a = \log a$

$\log a = 1$

$P \equiv (1, 1)$

focal distance of $\mathrm{P} = 1 + \frac{1}{4} = \frac{5}{4}$

Q.55 Answer:

40

Solution:

$(x - 1)\left(x^2 + 2x + 4\right) = 0$

roots are $\mathrm{a} = 1,\ \mathrm{b} = 2\omega \& \mathrm{r} = 2\omega^2$

$$\begin{vmatrix} 2 & 1 & 1 \\ 1 & -4 & 1 \\ 1 & 1 & -4 \end{vmatrix}$$

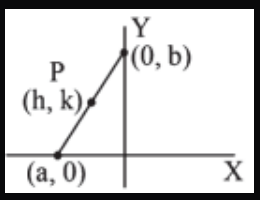
$2(16-1)-(-4-1)+(1+4)$

$30+5+5=40$

Q.56 Answer:

$2(x-y)+\ell=0$

Solution:



Length of x intercept $=-a$

Length of y intercept $=\mathrm{b}$

$\therefore$ Given $\mathrm{b}-\mathrm{a}=\ell$....(1)

Let circumcentre is $\mathrm{P}(\mathrm{h},\mathrm{k})$

$$\therefore \mathrm{h}=\frac{\mathrm{a}}{2},\mathrm{k}=\frac{\mathrm{b}}{2}$$

$2(\mathrm{k}-\mathrm{h})=\mathrm{b}-\mathrm{a}$

From (1) & (2) locus is

$2(y-x)=\ell$

Q.57 Answer:

2

Solution:

$x^2+y^2-2x-2y-2=0$

$\text{centre } \equiv(1,1),\ \text{radius } =2$

co-ordinate of a point on the curve

$\equiv(1+2\cos\theta,1+2\sin\theta)$

$$\Rightarrow \frac{8}{(x-1)^2}+\frac{(y-1)^2}{4}=\frac{2}{\cos^2\theta}+\sin^2\theta$$

using A.M. $\geq$ G.M.

min. value $=2$ at $\sin\theta=1$

Q.58 Answer:

1

Solution:

$x = 1, f(1) + 2f(1) = 1$

$\Rightarrow \quad f(1) = \frac{1}{3}$

$f\left(x^2\right) + 2x^2 f'\left(x^2\right) + \left(x^2+1\right) f'(x) + 2xf(x) = 1$

$x = 1$

$f(1) + 2f'(1) + 2f'(1) + 2f(1) = 1$

$4f'(1) + 1 = 1 \Rightarrow f'(1) = 0$

$2xf'\left(x^2\right) + 4x^3 f''\left(x^2\right) + 4xf'\left(x^2\right) + 2xf'(x)$

$+ \left(x^2+1\right) f''(x) + 2f(x) + 2f'(x) = 0$

$x = 1$

$2f'(1) + 4f''(1) + 4f'(1) + 2f'(1) + 2f''(1)$

$+ 2f(1) + 2f'(1) = 0$

$6f''(1) + \frac{2}{3} = 0$

$f''(1) = -\frac{1}{9} \Rightarrow 9f''(1) + 1 = 0$

Q.59 Answer:

$2\vec{a} - \vec{c} + 4\vec{b} = 0$

Solution:

$\overrightarrow{c} = \lambda\left(\frac{\overrightarrow{a}}{2} + \overrightarrow{b}\right)$

$16 = \lambda(2-1) \Rightarrow \lambda = 4$

Q.60 Answer:

Strictly increasing $\forall x \in R$

Solution:

Assume $f(x) = x^3 + x$ **so that** $g(x) = x^3; h(x) = x$ **and** $F(x) = x^3 - \frac{\sqrt{3}x^2}{2} + x$

$F'(x) = 3x^2 - \sqrt{3}x + 1$

$D = 3 - 12 < 0$

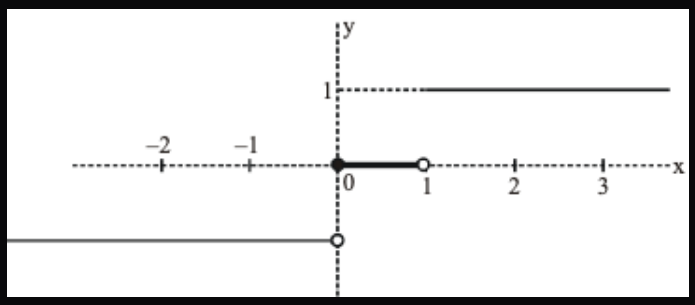
$F'(x) > 0 \forall x \in R$

$\Rightarrow F$ is increasing

Q.61 Answer:

Non differentiable at $x = 0$ & 1 only

Solution:



Q.62 Answer:

$\frac{1}{2}$

Solution:

$$f(\theta)=\frac{1-\sin 2\theta+\cos 2\theta}{2\cos 2\theta}=\frac{(\cos\theta-\sin\theta)^2+\left(\cos^2\theta-\sin^2\theta\right)}{2(\cos\theta-\sin\theta)(\cos\theta+\sin\theta)}=\frac{(\cos\theta-\sin\theta)+(\cos\theta+\sin\theta)}{2(\cos\theta+\sin\theta)}$$

$$=\frac{2\cos\theta}{2(\cos\theta+\sin\theta)}=\frac{1}{1+\tan\theta}$$

$$f\left(11^\circ\right)\cdot f\left(34^\circ\right)=\frac{1}{(1+\tan 11^\circ)}\cdot\frac{1}{(1+\tan 34^\circ)}=\frac{1}{(1+\tan 11^\circ)}\cdot\frac{1}{(1+\tan(45^\circ-11^\circ))}$$

$$=\frac{1}{(1+\tan 11^\circ)}\cdot\frac{1}{1+\frac{1-\tan 11^\circ}{1+\tan 11^\circ}}=\frac{1}{(1+\tan 11^\circ)}\cdot\frac{1+\tan 11^\circ}{2}=\frac{1}{2}$$

Q.63 Answer:

$\frac{3^{20}-13.2^{20}+43}{3^{20}}$

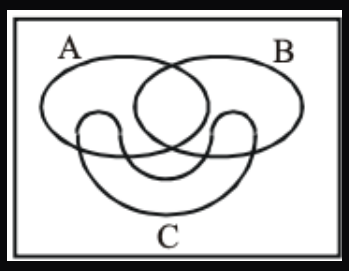
Solution:

$\mathrm{A}\rightarrow \mathrm{x}\leq 1$

$\mathrm{B}\rightarrow \mathrm{y}=0$

$\mathrm{C}\rightarrow \mathrm{z}=0$

$\mathrm{n}(\mathrm{A}\cup\mathrm{B}\cup\mathrm{C})$



$$=20\left(2^{19}\right)+1\left(2^{20}\right)+1\left(2^{20}\right)+1\left(2^{20}\right)$$

$$-((21\times 1))-1-(21\times 1)$$

Total cases $=3^{20}$

Ans. $=\frac{3^{20}-13.2^{20}+43}{3^{20}}$

Q.64 Answer:

$\frac{\pi}{2}-1$

Solution:

$\int_{\frac{-(\pi-2)}{4}}^{\frac{\pi-1}{2}} \left[\sin^{-1}(\sin(2x+1))\right] dx$

Let $2\mathrm{x}+1=\mathrm{t}$

$$\frac{1}{2}\int_{\frac{-\pi}{2}+2}^{\pi} \left[\sin^{-1}(\sin t)\right] \mathrm{dt}$$

$$=\frac{1}{2}\left[\int_{\frac{-\pi}{2}+2}^{\frac{\pi}{2}} [\mathrm{t}]\mathrm{dt}+\int_{\frac{\pi}{2}}^{\pi} [\pi-\mathrm{t}]\mathrm{dt}\right]$$

$$=\frac{1}{2}\left(\int_{\frac{-\pi}{2}+2}^{1} 0.\mathrm{dt}+\int_{1}^{\frac{\pi}{2}} 1\cdot\mathrm{dt}+\int_{\frac{\pi}{2}}^{\pi-1} 1\cdot\mathrm{dt}+\int_{\pi-1}^{\pi} 0.\mathrm{dt}\right)$$

$$=\frac{1}{2}\left(\left(\frac{\pi}{2}-1\right)+\frac{\pi}{2}-1\right)$$

$$=\frac{1}{2}[\pi-2]=\frac{\pi}{2}-1$$

Q.65 Answer:

697

Solution:

Let $z=a+ib$ where $b=164$

$$\therefore \quad \frac{a+ib}{(a+n)+ib}=4i$$

$$\Rightarrow a+ib=-4b+4(a+n)i$$

$\therefore$ On equating the real amd imaginary part, we get

$a=-4b=-4.164=-656$

and $b=4(a+n)\Rightarrow 164=4(a+n)\Rightarrow a+n=41\Rightarrow -656+n=41\Rightarrow n=656+41=697$

Q.66 Answer:

$[2,5)$

Solution:

$$f\{\mathrm{x}\}=\left\{\begin{array}{ll}\mathrm{x}, & \mathrm{x}<0\\ 1+\mathrm{x}^2, & \mathrm{x}\geq 0\end{array}\right\}$$

$\mathrm{g}(\mathrm{x})=(1+\{\mathrm{x}\})\geq 1\forall \mathrm{x}\in \mathrm{R}$

$f(\mathrm{g}(\mathrm{x}))=1+(1+\{\mathrm{x}\})^2\forall \mathrm{x}\in \mathrm{R}$

$0\leq\{\mathrm{x}\}<1$

$1\leq 1+\{\mathrm{x}\}+2$

$1\leq(1+\{\mathrm{x}\})^2<4$

$2\leq 1+(1+\{\mathrm{x}\})^2<5$

Range of $f(g(x))=[2,5)$

Q.67 Answer:

80

Solution:

First element in A_{10} is $a_{1^2+2^2+3^2+\ldots\ldots\ldots\ldots 9^2+1} = a_{286}$ and $[a_{286}] = 8$ and last element is a_{385}

$a_{385} = [\log_2 385] = 8$

$\text{Sum } = 80$

Q.68 Answer:

1

Solution:

$$f'(1^+) = \text{Lim}_{h\to 0}\frac{2(1+h)+b-(2+b)}{h} = 2$$

$$f'(1^-) = \text{Lim}_{h\to 0}\frac{(1-h)^2+a-(2+b)}{-h} = \text{Lim}_{h\to 0}\frac{(h^2-2h)+a-b-1}{-h} = 2$$

where $a-b-1=0$ ...(1)

$g'(1^-) = \text{Lim}_{h\to 0}\frac{3(1-h)+b-1}{-h} = 3$

where $2+b=0 \Rightarrow b=-2$ ...(2)

$g'(1^+) = \text{Lim}_{h\to 0}\frac{(1+h)^3-1}{h} = \text{Lim}_{h\to 0}\frac{1+3h+3h^2+h^3-1}{h} = 3$

from (1) and (2) $a=-1$; $\frac{f'(x)}{g'(x)}$ at $x=1$

Q.69 Answer:

π

Solution:

$$I = \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{x^4}{1-x^4}\cos^{-1}\frac{2x}{1+x^2}dx \quad \ldots(1)$$

$$I = \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{x^4}{1-x^4}\cos^{-1}\left(\frac{-2x}{1-x^4}\right)dx \text{ (using King)}$$

$$I = \int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{x^4}{1-x^4}\left(\pi-\cos^{-1}\frac{2x}{1-x^4}\right)dx \quad \ldots(2)$$

add (1) and (2)

$$\therefore 2I = \pi\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{x^4}{1-x^4}dx$$

$$2\text{I } = 2\pi\int_{0}^{1/\sqrt{3}} \frac{x^4}{1-x^4}dx$$

$\therefore \quad \text{k} = \pi$ Ans.

Q.70 Answer:

0

Solution:

Point on the line $= (2t, 2+3t, 3+4t)$

Equating to $B(q,5,7)$ we get $t=1$ and hence $q=2$

$\therefore B = (2,5,7)$

D.R.s of AB are $= (p-1,-6,4)$

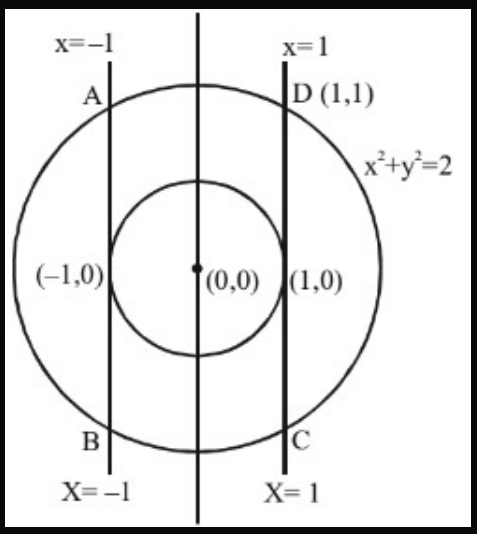
$\text{AB} \perp \text{to the line} \Rightarrow (p-1)(2)+(-6)(3)+(4)(4)=0$

$\therefore p=2 \text{ and } p-q=0$

Q.71 Answer:

—

Solution:



Given curve $x^2y^2+x^4-3x^2-y^2+2=0$

$\Rightarrow \quad \left(x^2-1\right)\left(x^2+y^2-2\right)=0$

$$\text{bounded area } = 4\int_0^1 \sqrt{2-x^2}$$

$$= 4\left[\left[\frac{x}{2}\sqrt{2-x^2}\right]_0^1+\frac{2}{2}\left[\sin^{-1}\frac{x}{\sqrt{2}}\right]_0^1\right]$$

$$= 4\left(\frac{1}{2}+\frac{\pi}{4}\right)=2+\pi$$

Q.72 Answer:

—

Solution:

$$\overline{x}=\frac{\sum x_i}{100}=0\&$$

$$\frac{\sum |x_i-\overline{x}|}{100}=5 \Rightarrow \sum |x_i|=500$$

$$\Rightarrow \sum x_i^2+2\sum_{1\le i<j\le 100}|x_ix_j|=(500)^2$$

$$\Rightarrow \frac{\sum x_i^2}{100}=\frac{(500)^2-2\sum |x_ix_j|}{100}=2500-1600$$

$$\text{S.D. } = \sqrt{\frac{\sum (x_i-\overline{x})^2}{100}}=\sqrt{900}=30$$

Q.73 Answer:

—

Solution:

Let $f(x) = (1+2x)^n$.
Sum of coefficients is given by $f(1)$.
So, we have $(1+2)^n = 6561 \Rightarrow 3^n = 6561 \Rightarrow n = 8$.
Thus, number of terms in the expansion of $(1+2x)^8$ is 9 .
Hence, the greatest term for $x = \frac{1}{2}$ is the 5th term since $2x = 1$.
Thus, $k^k = 5^5 = 3125$

Q.74 Answer:

—

Solution:

$$\text{Let } \frac{x^2+y^2}{x^2-y^2} = t$$
$$\Rightarrow t + \frac{1}{t} = 2$$
$$\Rightarrow (t-1)^2 = 0$$
$$\Rightarrow t = 1$$
$$\Rightarrow x^2 + y^2 = x^2 - y^2 \Rightarrow 2y^2 = 0$$
$$\Rightarrow y = 0$$
$$\text{Put in } \frac{x^8+y^8}{x^8-y^8} + \frac{x^8-y^8}{x^8+y^8}, \text{ we get Ans. } = 2$$

Q.75 Answer:

—

Solution:

$$f'(x) = f(x) \cdot n(x+1)^{n-1}$$
$$f''(x) = f(x)n(n-1)(x+1)^{n-2} + f^1(x)n(x+1)^{n-1}$$
$$f''(1) = n(n-1)f(1)2^{n-2} + f(1)n2^{n-1} \cdot n2^{n-1}$$
$$= 67.2^n \cdot e^{2^n}$$
$$n \cdot 2^{n-2}\,(n-1+n\cdot 2^n) = 67.2^n$$
$$\frac{n}{4}\,(n-1+n\cdot 2^n) = 67 \quad \Rightarrow n = 4$$